

Phloem metabolites and seed composition

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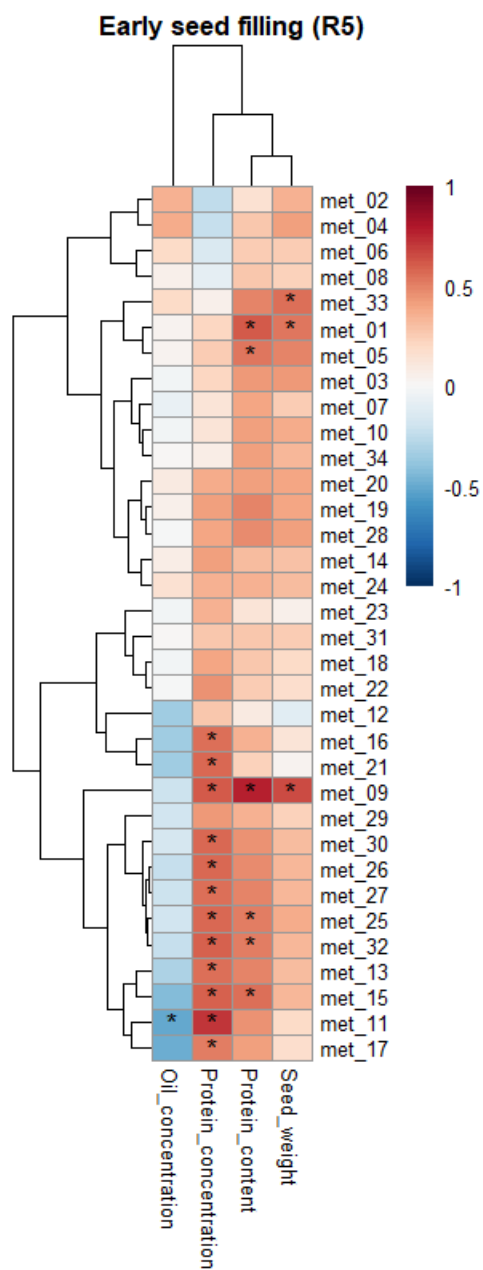
Biological background

Phloem-delivered metabolites were measured in six contrasting soybean genotypes at two developmental stages (R5 and R6). Pearson correlations were calculated between phloem metabolite levels and final seed composition (protein content, protein concentration, oil concentration, seed weight).

Results

Early seed filling (R5)

metabolite	composition_var	r	p	abs_r
met_11	Oil_concentration	-0.5099202	0.0306278	0.5099202
met_17	Oil_concentration	-0.4934421	0.0374355	0.4934421
met_11	Protein_concentration	0.7042599	0.0011039	0.7042599
met_09	Protein_concentration	0.6006076	0.0083959	0.6006076
met_15	Protein_concentration	0.5998491	0.0085004	0.5998491
met_32	Protein_concentration	0.5904943	0.0098776	0.5904943
met_25	Protein_concentration	0.5766655	0.0122358	0.5766655
met_09	Protein_content	0.7619911	0.0002376	0.7619911
met_01	Protein_content	0.6014610	0.0082795	0.6014610
met_15	Protein_content	0.5496969	0.0181172	0.5496969
met_05	Protein_content	0.5343685	0.0223395	0.5343685
met_25	Protein_content	0.5160816	0.0283447	0.5160816
met_09	Seed_weight	0.6523641	0.0033427	0.6523641
met_33	Seed_weight	0.5425861	0.0199896	0.5425861
met_01	Seed_weight	0.5392788	0.0209108	0.5392788
met_05	Seed_weight	0.4945696	0.0369356	0.4945696



Mid-Stage seed filling (R6)

metabolite	composition_var	r	p	abs_r
met_30	Oil_concentration	-0.4825902	0.0425157	0.4825902
met_15	Protein_concentration	0.6738183	0.0021692	0.6738183
met_21	Protein_concentration	0.6144973	0.0066581	0.6144973
met_26	Protein_concentration	0.6080936	0.0074189	0.6080936
met_27	Protein_concentration	0.6054476	0.0077531	0.6054476
met_12	Protein_concentration	0.5975694	0.0088208	0.5975694
met_17	Protein_content	0.7274079	0.0006239	0.7274079
met_27	Protein_content	0.7023939	0.0011532	0.7023939
met_32	Protein_content	0.6752665	0.0021043	0.6752665
met_16	Protein_content	0.6706553	0.0023169	0.6706553
met_31	Protein_content	0.6440514	0.0039185	0.6440514
met_34	Seed_weight	0.6834315	0.0017672	0.6834315
met_17	Seed_weight	0.6660923	0.0025445	0.6660923
met_05	Seed_weight	0.6564329	0.0030873	0.6564329
met_13	Seed_weight	0.6555644	0.0031404	0.6555644
met_14	Seed_weight	0.6399994	0.0042273	0.6399994

Mid-Stage seed filling (R6)

